

GUHAN VENKAT

guhan.venkatachalapathi@gmail.com | 201-424-2943 | Skillman, NJ | Age: 15

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, PyTorch, FastAPI, Next.js, React, NumPy, SciPy

ML / AI: V-JEPA2, BioBERT, ResNet, Vision Transformers, CLIP, YOLOv8, LSTM, fine-tuning

Computer Vision: MediaPipe, SLEAP, OpenCV, FACS action unit detection, landmark regression, pose estimation

Neuroscience Tools: fMRI cortical encoding, fsaverage5 atlas, Glasser-Yeo ROI projection, behavioral phenotyping pipelines

Infra & APIs: HuggingFace, Azure Cognitive Services, OpenAI API, Anthropic API, WebSocket, CUDA, Git

RESEARCH EXPERIENCE

Research Assistant | Eisch Lab, UPenn / CHOP

Jan 2026 - Present

- **ROI Analysis Software:** Wrote a no-code GUI that takes raw SLEAP body-part pixel predictions and runs them through a configurable ROI analysis pipeline, cutting out manual annotation entirely.
- **Behavioral tracking pipeline:** Wired together interaction scoring, locomotion speed, and head-direction tracking into one real-time pipeline, replacing hand-coded annotation across hippocampal dentate gyrus behavior experiments.

Research Engineer | UPenn

Jan 2026 - Present

- **Zebrafish ID Model:** Training a deep learning model to tell apart **80,000+** individual zebrafish by their stripe patterns, making it possible to track the same fish across weeks of vascular development experiments.
- **Local data pipeline:** Built an on-device imaging and labeling system to handle the volume of zebrafish data the model needed, with no cloud dependency.

Research Assistant | Mukherjee Lab, CHOP / UPenn

Feb 2026 - Present

- **Schizophrenia phenotyping model:** Building a model that reads fine-grained facial landmarks from mice to find expression patterns tied to schizophrenia-related emotional states.
- **Circuit mapping pipeline:** Connecting those expression predictions to prefrontal-hippocampal circuit disruptions seen in mouse models of neurodevelopmental disorders, aimed at informing psychiatric research.

ML Intern | Presonance

Apr 2026 - Present

- **Neural prediction pipeline:** Training and optimizing the core ML model that simulates human brain response to short-form video content, achieving 0.827 AUC in top vs. bottom quartile viral prediction across 437 real-world reels.
- **Model optimization:** Improving inference speed and prediction accuracy on the frame-level engagement and buyer intent models, enabling per-second brain-state diagnostics across audience segments and content formats.

TECHNICAL PROJECTS

RIBOSCOPE: Mechanistic Interpretability of RNA Foundation Models | PyTorch | BatchTopK SAE | Cross-layer Transcoders | RNA-FM | ErnieRNA | RNA-MSM | OmniGenome | CUDA | ISEF 2027

- **Interpretability framework:** Building the first mechanistic interpretability framework for RNA foundation models: training BatchTopK SAEs and cross-layer transcoders across 4 architecturally distinct models (RNA-FM, ErnieRNA, RNA-MSM, OmniGenome) to recover hundreds of biological circuits learned without any labeled training data.
- **Cross-model discovery:** Trained SAEs on RNA-FM (EV=0.882) and ErnieRNA (EV=0.674), recovering 84 reciprocal-match feature pairs at Jaccard ≥ 0.5 ; confirmed motifs span tRNA T Ψ C loops, U2/U6 snRNAs, snoRNA C/D-box, and riboswitch aptamer scaffolds, producing ground-truth-like biological signal with zero labeled training data.
- **Disease anchor + validation:** Directly addresses the open circularity critique in bio-interpretability by grounding every feature against MPRA and DMS-MaPseq experimental datasets, validated across three disease-linked lncRNAs: MALAT1 (lung cancer metastasis), NEAT1 (Alzheimer's/ALS/SARS-CoV-2), and BACE1-AS (amyloid regulation). Targeting Nature Methods and ISEF 2027 CBIO.

EEG Seizure Prediction - Domain Gap Study | PyTorch | CNN | Transformer | GNN | CHB-MIT | Siena | Frontiers in Neuroinformatics

- **Cross-cohort generalization:** Trained seizure prediction models on pediatric CHB-MIT EEG (23 patients, 74,658 segments) and zero-shot evaluated on adult Siena EEG - no target data used at any stage.
- **Central finding:** Normalization strategy - not model architecture - is the primary driver of cross-dataset generalization failure. Replacing BatchNorm with InstanceNorm recovered zero-shot F1 from 0.025 to 0.281, matching domain adversarial methods that require target data (Wilcoxon $p=0.0005$, $r=-0.97$).
- **Model comparison:** Benchmarked CNN, Transformer, and GNN architectures bidirectionally; characterized an asymmetric degradation pattern showing attention-based models are more data-hungry than CNNs under distribution shift. Targeting Frontiers in Neuroinformatics and ISEF 2027.

NEXUS- α : Multi-Modal Frontier Trading Model | PyTorch | Mamba / SSM | iTransformer | Chronos | VisionTS | FinGPT | Sparse MoE | RLMF

- **Novel architecture:** Designed a tri-modal price tokenization pipeline (quantile bins via Chronos, image patches via VisionTS, float embeddings) fused via cross-attention, fed into an Inverted-Mamba backbone stacking Mamba SSM for temporal modeling with iTransformer variate-axis attention for cross-asset correlation, achieving linear complexity in both time and variates.
- **End-to-end training pipeline:** Implemented a 3-tier hierarchical sparse MoE router (macro regime, meso pattern, micro signal) with TFT-style gated modality weighting across price, sentiment, macro, and order book inputs; trained a DQN execution agent via reinforcement learning from market feedback (RLMF) with gradients flowing back into the forecasting backbone, optimizing a unified loss over quantile NLL, regime classification, Sharpe, and RLMF reward.
- **Paper trading results:** Achieved **36.2% simulated annualized return** on NQ futures (Sharpe 2.11, win rate 63.4%, max drawdown 8.3%), generating **+11.4% alpha** over the NQ benchmark on a risk-adjusted basis over a 6-month simulated window.

Neural Attention Analysis Dashboard | PyTorch | V-JEPA2 | YOLOv8 | CLIP | FastAPI | Next.js | CUDA

- **Brain encoding pipeline:** Runs a state-of-the-art cortical encoding model on short-form video to predict fMRI activation across all **20,484 cortical vertices** per timepoint, with V-JEPA2 providing spatiotemporal video features as input.
- **Neural engagement signal:** Derives a per-timepoint engagement score from the push-pull between external-drive networks (visual, motion, semantic, frontoparietal attention) and the default mode network - a continuous brain-state signal extracted from predicted cortical activation.
- **Visual-neural correlation:** Runs YOLOv8 and CLIP in parallel to identify and label on-screen objects per frame, then correlates object presence with predicted activation across **5 Glasser-Yeo functional ROI groups** - revealing which visual content drives or suppresses neural engagement.

ToneSpeech | MediaPipe | Python | Azure Neural TTS | SSML | OpenAI API | PyQt6 | VB-Cable

- **ASL-to-speech pipeline:** Desktop app that translates ASL into emotionally expressive spoken audio in real time and routes it as a virtual microphone into Zoom, Teams, and Google Meet with no screen sharing or captions.
- **Action Unit detector:** Wrote a FACS-grounded AU detector from scratch using MediaPipe Face Mesh (**468 landmarks**), computing geometric displacement ratios to output emotion labels and intensity scores between 0 and 1, no external emotion model.
- **Full synthesis stack:** Handles the full path from AU vector to LLM emotional reconstruction, SSML prosody tagging, and Azure Neural TTS. Built to capture not just what was signed, but how.

Clinical Trial Outcome Predictor | BioBERT | Python | Google Gemini | Gradio | ClinicalTrials.gov API

- **Outcome forecasting:** Fine-tuned BioBERT (110M parameters) on **472 completed clinical trials** (Phase 2 and 3, sourced from ClinicalTrials.gov) to predict trial success or failure from protocol descriptions, achieving **84.5% accuracy, 91.6% F1, and 97.5% recall** across cancer, cardiovascular, neurological, and respiratory conditions.
- **Explainability layer:** Integrated Google Gemini to generate natural language explanations of each prediction including confidence scores and risk factors, delivered through a real-time Gradio web interface with follow-up question support.

LEADERSHIP

Event Lead | Relay for Life, Bridgewater, NJ

Feb 2025 - Present

- **Fundraising:** Raised **approximately \$56,000** for cancer research by writing sponsorship proposals, securing donors, and running the full fundraising pipeline for a large community event.

- **Team and event management:** Managed volunteers, booth logistics, and day-of operations; also put together cancer awareness programming for attendees.

Financial Officer, Incoming President | NexGen STEM

Sept 2024 - Present

- **Grant writing:** Wrote grant proposals that paid for **1,500+ STEM kits** sent to underserved kids, while managing the budget and logistics across several simultaneous projects.
- **International program:** Led a trip to Senegal to teach students water purification science; taking over as sole President in September 2026.

EDUCATION

Somerset County Vocational and Technical School | Academy of Health and Medical Sciences *GPA: 4.0 UW* |
Expected Jun 2028

Awards: Rutgers High School Science Symposium Finalist (AI/oncology) | Entrepreneurship State Competition, 2nd Place | Basic Life Safety Certified (CPR/AED)